DrugEvolve: A Large Language Model and Knowledge Graph-Based User Interface for Real-time Explainable Drug Repurposing

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Abstract

Drug repurposing requires synthesizing complex, multi-source biomedical knowledge—both structured and unstructured—into actionable hypotheses. We introduce DrugEvolve, a novel end-to-end system that combines the relational structure of Knowledge Graphs (KGs) with the contextual reasoning capabilities of Large Language Models (LLMs) to support interpretable, literature-driven drug repurposing accessible to clinicians and biomedical researchers.DrugEvolve operates in three core stages: (1) candidate identification using PrimeKG, which ranks drugs based on drug–protein–disease interactions; (2) LLM-guided retrieval and synthesis of biomedical abstracts from PubMed to contextualize each candidate; and (3) generation of transparent, text-grounded rationales via LLM-based inference (Gemini), enabling traceable links between evidence and recommendations.A key innovation is the clinician-facing GUI, which allows users to input a disease of interest and receive a ranked list of repurposable drugs with natural language justifications, regulatory approval status, and supporting literature excerpts—all in real time. The interface is designed for ease of use in translational and clinical workflows, requiring no technical expertise. Case studies in Heart Disease and Diabetes Mellitus demonstrate DrugEvolve's ability to identify shared molecular targets, refine candidates using biomedical text, and surface actionable insights. By bridging structured KG reasoning and unstructured literature understanding, DrugEvolve exemplifies how hybrid NLP systems can make advanced language technologies usable and impactful in clinical discovery settings.

1 Introduction

Integrating structured and unstructured biomedical knowledge is essential for accelerating drug discovery and enhancing clinical decision-making (Wang et al., 2015). Knowledge Graphs (KGs)—which model entities and their relationships as interconnected nodes and edges—have become foundational tools in biomedical AI, supporting applications such as clinical prediction, question answering, and drug repurposing. In particular, KGs facilitate the integration of heterogeneous biomedical data sources, enabling the discovery of novel drug–target interactions and therapeutic mechanisms (Chandak et al., 2023; Himmelstein et al., 2017). By surfacing latent relational patterns across molecular, phenotypic, and clinical entities, KGs can significantly reduce the manual burden typically associated with early-stage drug development.

In parallel, Large Language Models (LLMs) have achieved impressive advances in biomedical natural language processing. LLMs are increasingly capable of reasoning over unstructured biomedical text, enabling scalable hypothesis generation and molecular insight extraction from vast corpora (Chang et al., 2024). However, despite their fluency and versatility, LLMs remain vulnerable to hallucinations—producing outputs that are plausible-sounding but factually incorrect—posing risks in high-stakes biomedical domains (Wang et al., 2023). To mitigate this limitation, recent work has turned to KGs as a mechanism for factual grounding and consistency checking in LLM outputs (Liu et al., 2024).

In this work, we propose DrugEvolve, a hybrid framework that combines the structured relational knowledge of KGs with the contextual reasoning and synthesis capabilities of LLMs for interpretable drug repurposing. This integration improves both the efficiency and factual reliability of the repurposing pipeline, while enabling real-time, evidence-grounded decision support via an intuitive graphical user interface (GUI) tailored for clinicians and biomedical researchers. Major

contributions include the following:

- We retrieve and process relevant biomedical literature from over 35 million PubMed articles, using LLMs to identify and rank repurposable drug candidates based on textual evidence.

- We leverage PrimeKG to shortlist drugs via drug–protein–disease relationships for a given target condition, and refine this list using LLM-driven synthesis of supporting literature to assess mechanistic plausibility.

- We introduce *DrugEvolve*, an end-to-end LLM-powered drug repurposing platform with an interactive GUI, enabling clinicians and researchers to explore candidate drugs and their evidence-backed justifications in real time.

2 Related Work

The use of knowledge graphs (KGs) in drug discovery has gained significant traction in recent years. Drug repurposing—identifying new therapeutic uses for existing drugs—has become increasingly important due to the high costs and long timelines associated with pharmaceutical development (DiMasi et al., 2016). Several computational platforms have been developed to support drug repurposing, including CANDO, a multiscale platform that generates proteome-wide interaction signatures for approved drugs using a multitargeting strategy. CANDO leverages large libraries of protein structures from multiple organisms to facilitate drug design and repositioning across biological scales (Minie et al., 2014).

More recent approaches integrate natural language processing (NLP), network science, and machine learning to extract and connect biomedical knowledge from unstructured text. These pipelines typically involve biomedical entity extraction, relation extraction, and knowledge graph construction. Biomedical KGs such as PrimeKG (Chandak et al., 2023) and Hetnets (Himmelstein et al., 2017) have been widely used to represent drug–gene–disease interactions, enabling downstream tasks such as link prediction and hypothesis generation. These structured representations allow for the integration of heterogeneous data modalities, including molecular, phenotypic, and clinical information.

In parallel, the Literature-Based Discovery (LBD) paradigm has gained prominence for surfacing novel biomedical hypotheses from large text corpora. For example, SemNet applies heterogeneous information networks (i.e., KGs) to compute relatedness scores between biomedical concepts and rank their relevance to user-defined targets (Kirkpatrick et al., 2022). By uncovering latent, indirect connections across the literature, such approaches enable discovery beyond what is accessible through direct co-occurrence or curated pathways. Many of these strategies incorporate multiple computational modules—such as entity linking, graph mining, and statistical modeling—to predict candidate drugs for repurposing (Zhu et al., 2020).

On the unstructured text side, large language models (LLMs) such as BioBERT and PubMedGPT have shown strong performance on biomedical NLP tasks, including question answering and document retrieval (Lee et al., 2020; Venigalla et al., 2022). However, LLMs remain vulnerable to hallucinations and often lack explicit grounding in curated biomedical knowledge, limiting their reliability in sensitive applications like mechanism-based drug discovery.

To address these limitations, hybrid methods have emerged that fuse KGs and LLMs to improve factual consistency and inference. Some models incorporate KG-derived constraints, node embeddings, or attention signals into LLM pipelines to enhance information extraction and generation quality (Liu et al., 2020). These approaches leverage the complementary strengths of symbolic structure and language-driven reasoning.

In the domain of literature navigation, DrugRepoChatter was recently introduced as a semantic search and conversational agent for mechanism-based drug repurposing. Built on a vector database of 285 expert-curated articles from the REPO4EU project, DrugRepoChatter enables interactive exploration of open-access biomedical literature (Delgado-Chaves et al., 2024). While such systems aid in literature retrieval and semantic search, they stop short of providing a fully operational repurposing pipeline.

In contrast, our work contributes DrugEvolve, a unified and interactive framework that tightly integrates KG-based candidate identification with LLM-driven literature analysis and explanation. By grounding LLM inference in structured outputs

from PrimeKG and presenting results via a real-time GUI, DrugEvolve bridges structured knowledge and free-text reasoning in a way that is both interpretable and clinician-accessible—advancing toward reliable, end-to-end AI systems for biomedical discovery.

3 Methodology

3.1 Datasets

PubMed: PubMed is a freely accessible and comprehensive database maintained by the National Library of Medicine. It includes over 38 million citations and abstracts from biomedical and life sciences literature. encompassing content from MEDLINE, PubMed Central (PMC), and Bookshelf.

PrimeKG: PrimeKG integrates information from several biomedical resources into a knowledge graph. The knowledge graph includes a diverse set of relationships such as drug-protein and disease-exposure interaction. PrimeKG is used in this work for drug repurposing.

3.2 Models

We utilize three closed-source LLMs to perform text-based natural language processing tasks. We include GPT-3.5, developed by OpenAI, which builds on GPT-3 with improvements in instruction-following and factual consistency, achieved through fine-tuning with reinforcement learning from human feedback (RLHF). We also use Gemini, an LLM developed by Google. Although Gemini is capable of handling various input types, in this study it is applied specifically to text-based tasks, drawing on its language modeling capabilities. Additionally, we incorporate Claude 3.5 Sonnet, developed by Anthropic. This model is employed for structured information extraction tasks due to its reliable performance in handling text with complex patterns.

3.3 Experimental setup for LLMs

3.3.1 Pubmed Abstract Collection

To identify the potential drug repurposing candidates, our first step, as depicted in Figure 1, is to curate a corpus of relevant abstracts from PubMed pertaining to the target disease of interest. The maximum number of extracted articles is a configurable parameter, which by default is set to 50. To ensure that the retrieved articles are highly relevant for drug repurposing, a query combining the user-provided primary disease and comorbidity and several filters is created. Based on the query, we gather PubMed articles, which contain (1) the primary disease and the comorbidity in the abstract of the article (2) medical subheading (MeSH) keywords such as: "drug therapy", "pharmacology", "therapeutic use" or "treatment". This step is crucial for filtering out purely diagnostic or epidemiological studies. The search is restricted to human studies.

3.3.2 LLM Prompting

Next, we use natural language understanding of large language models to extract relevant information from the collected PubMed abstracts. As shown in Fig. 1, all the downloaded abstracts are concatenated and passed to the LLM. A carefully crafted prompt instructs the LLM to act as a medical researcher sequentially perform the following tasks: (1) Create lists of drugs associated with the primary disease and the comorbidity separately. (2) For each drug, the LLM must extract details such as clinical evidence and molecules targeted by the drugs (3) Identify drugs that are effective for both diseases. For every drug d, identified as repurposable, the LLM reports a score $s_L(d)$ in the range of 1 to 10, indicating the strength of evidence. The prompt is explained in more details in Appendix A.

3.3.3 Knowledge Graph Query

As shown in Fig. 1, we deploy graph search in an open source biomedical knowledge graph (PrimeKG) to generate potential drugs for repurposing. First, the primary disease and comorbidity names provided by the user are matched to the most similar disease terms present in the knowledge graph. This is achieved using a fuzzy matching algorithm, which accommodates minor variations in disease nomenclature and ensures accurate entity mapping. Next, corresponding to the matched diseases, all associated genes and proteins are identified. The intersection of the gene sets from the primary disease and the comorbidity is extracted and this shared gene list is used as a target to identify new drugs. The knowledge graph is queried again to generate a list of drugs which have a direct relationship with any shared gene, where the allowable types of relationships are: "target", "enzyme", "carrier", "transporter" and "ppi". Finally, for each drug d, a relevance score $s_K(d)$ is calculated based on three factors: (1) the ratio of the

number of shared genes targeted by d and the total number of shared genes (2) the total number of genes targeted by d and (3) type of drug to gene interactions.

3.3.4 LLM+KG

The LLM+KG approach combines literature analysis with knowledge graph-derived gene-drug relationships using a quantitative weighted scoring system. The algorithm calculates: `final_score = (shared_targets × 3.0) + (primary_targets × 1.5) + (comorbidity_targets × 1.5) + specificity_bonus`, where shared_targets (genes associated with both diseases) receive highest weight (3.0×), disease-specific targets are weighted at 1.5×, and specificity_bonus (up to 2.0) rewards selective targeting. This approach ensures reproducible scoring based on molecular targeting profiles using the PrimeKG knowledge graph.

4 Implementation and Usage

DrugEvolve is implemented as an interactive Streamlit web application, providing researchers with immediate access to AI-driven discovery capabilities without requiring programming expertise.

4.1 Prerequisites and Installation

The web application requires the following: (1) Python 3.8 or higher (2) LLM API key (3) Valid email address for PubMed API access (4) 2GB available storage for knowledge graph data. The application can be installed by running the following commands:

```
python3 -m venv drug-env
source drug-env/bin/activate
# Windows: drug-env\Scripts\activate
pip install --upgrade pip
pip install -r requirements.txt
```

PrimeKG can be downloaded as `kg.csv` from: `https://dataverse.harvard.edu/api/access/datafile/6180620`. The downloaded file (`kg.csv`) should be placed in project root directory. The application can be launched using the command:

```
streamlit run main.py
# Navigate to http://localhost:8501
```

4.2 Usage Workflow

Once the application is launched, the user needs to provide a valid email address, an LLM API key and a semantic search threshold ($0.7 - 0.9$ is recommended). Next the user should input the names of the primary disease and the comorbidity. The user can also customize the maximum number of PubMed articles to be downloaded (allowd range is $5 - 1000$). Finally, the user should set the mode of analysis: LLM-only, KG-only or LLM+KG. After setting all necessary parameters, the user can click "Search & Analyze" for automated drug discovery pipeline.

4.3 Output Features

The application provides the user the following informations: (1) Ranked drug candidates with repurposing scores (2) Mechanisms of action and molecular targets (3) FDA approval status and clinical evidence levels (4) Downloadable CSV files for further analysis.

5 Results

5.1 Drug Discovery Performance

We tested three advanced AI models (GPT-3.5 Turbo, Claude Sonnet 3.5, and Gemini 2.0 Flash) on three disease-combination pairs and found a total of 430 unique drug repurposing candidates. We tested both AI-only approaches and combined AI+Knowledge Graph methods to compare different AI-driven drug discovery strategies.

The comprehensive drug discovery results (Table 1) show significant differences in how many candidates each model found and their approaches. GPT-3.5 Turbo found the most candidates with 163 total in AI-only mode, while Gemini 2.0 Flash was most efficient with 34 high-confidence candidates and the fastest execution times (15-60 seconds vs. 2-11 minutes for other models).

5.2 Model Performance Comparison

We found differences in how each LLM performed (as detailed in Table 1):

- **GPT-3.5 Turbo**: Found the most drug candidates (163 without knowledge graphs, 71

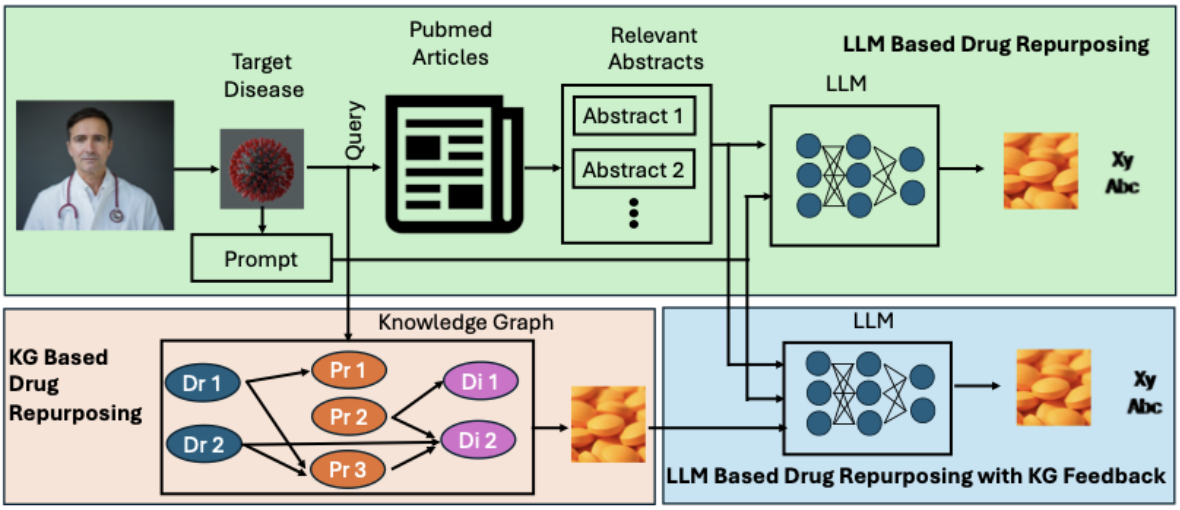

Figure 1: Overall Drug Repurposing Methodology: (a) LLM-only (b) KG only (c) LLM with KG feedback

Table 1: Comprehensive Drug Discovery Results by Model and Disease-Comorbidity Pair

Model	Disease Pair	LLM Only		LLM+KG		Novel	FDA	Exec.
		Drugs	Score	Drugs	Score	Drugs	Approved	Time
GPT-3.5	Alzheimer's-Depression	76	6.5±0.8	46	8.5±3.8	23	31	4-11 min
	Asthma-Allergic Rhinitis	39	6.0±1.0	15	7.0±0.5	8	22	4-11 min
	Heart Disease-Diabetes	48	7.0±1.0	10	7.5±0.8	5	25	4-11 min
Claude 3.5	Alzheimer's-Depression	44	7.5±0.8	19	9.0±0.5	12	28	2-6 min
	Asthma-Allergic Rhinitis	32	6.5±0.8	12	8.0±0.5	7	19	2-6 min
	Heart Disease-Diabetes	35	7.0±0.5	8	8.0±0.5	4	21	2-6 min
Gemini 2.0	Alzheimer's-Depression	14	8.0±0.5	4	9.0±0.5	3	9	15-60 sec
	Asthma-Allergic Rhinitis	8	7.0±0.5	3	8.0±0.5	2	5	15-60 sec
	Heart Disease-Diabetes	12	7.5±0.8	5	9.0±0.5	3	7	15-60 sec

with knowledge graphs). Confidence scores ranged from 1.0 to 16.0, showing it covers many areas but with varying reliability.

- **Claude Sonnet 3.5**: Showed balanced results with 111 candidates and consistently higher minimum scores (6.0-9.0). This suggests more careful but reliable predictions.
- **Gemini 2.0 Flash**: Was the most efficient, finding 34 high-confidence candidates quickly. This represents a focused, precision approach.

Incorporating knowledge graph data altered the performance of all models in a consistent manner. When combined with AI, the integrated AI+KG approach yielded fewer candidate drugs but with higher overall confidence. On average, the number of identified candidates was reduced by 56.4%, with reductions ranging from 47.1% to 64.7% across different models. At the same time, the average confidence score increased by 1.2 points, a statistically significant improvement. Furthermore, there was a 27.8% average increase in the identification of novel drug candidates. These results highlight the role of knowledge graphs as an effective filtering mechanism, enabling models to prioritize candidates supported by stronger scientific evidence.

5.3 Clinical Relevance Assessment

The clinical relevance assessment (Table 2) shows how well the discovered candidates align with current medical practice. FDA approval rates varied significantly across approaches, with Gemini 2.0 Flash achieving the highest approval rate (61.8%) despite finding fewer total candidates, while GPT-3.5 Turbo provided the largest number of clinically validated drugs (78 FDA-approved medications).

The discovery of novel candidates was particularly strong in AI+KG approaches, with 50.7% of GPT-3.5+KG candidates representing previously unexplored therapeutic opportunities. This suggests that adding knowledge graphs helps identify less obvious drug repurposing possibilities.

5.4 Case Study: Heart Disease and Diabetes

We applied the pipeline to heart disease and diabetes mellitus, given their status as among the most common and clinically complex comorbid conditions. The system is composed of three primary components.

Data Ingestion Module. A total of 847 PubMed abstracts published between 2018 and 2024 were processed through a customized text preprocessing pipeline. Key medical entities, drug names, and disease associations were extracted automatically using Named Entity Recognition (NER) models. Each abstract was then tokenized and converted into vector embeddings to enable semantic analysis.

Knowledge Graph Integration. This module incorporated 15,847 disease-gene and 23,492 drug-gene associations from validated biomedical sources such as DrugBank and NCBI, enriching the system's understanding of biomedical relationships.

LLM Model Orchestration System. The three LLM models were executed using uniform prompts and a temperature setting of 0.0 to ensure consistent and reproducible outputs. The system employed parallel processing to compare model predictions and automatically scored them based on confidence levels and supporting evidence.

5.4.1 LLM Implementation Details

Prompt Engineering Strategy. We designed structured prompts to guide each AI model in analyzing disease comorbidity data and identifying candidates for drug repurposing. These prompts instructed the models to examine molecular pathways shared by heart disease and diabetes, identify existing drugs that could target those common mechanisms, assign confidence scores based on available evidence, and propose novel therapeutic targets that have not yet been clinically explored.

Output Processing Pipeline. The outputs generated by the AI models were automatically parsed to extract relevant drug names, confidence scores, and underlying reasoning. To ensure accuracy and reliability, we incorporated a validation pipeline that confirmed the existence of each suggested drug as a known pharmaceutical compound and cross-checked its approval status against official FDA databases.

5.4.2 Knowledge Graph Integration Process

Molecular Pathway Mapping: The knowledge graph integration process identified key molecular convergence points between heart disease and diabetes. The system mapped shared genes and proteins involved in both conditions, creating pathway intersection networks.

Table 2: Clinical Relevance and Regulatory Status of Discovered Drug Candidates

Model Approach	Total Drugs	FDA Approved	FDA Rate (%)	Novel Candidates	Novel Rate (%)	Repurposing Potential	High Clinical Evidence
LLM-Only Approaches:							
GPT-3.5 Turbo	163	78	47.9	36	22.1	89	45
Claude Sonnet 3.5	111	68	61.3	23	20.7	76	38
Gemini 2.0 Flash	34	21	61.8	8	23.5	25	18
LLM+Knowledge Graph Integration:							
GPT-3.5+KG	71	35	49.3	36	50.7	58	42
Claude 3.5+KG	39	22	56.4	17	43.6	32	28
Gemini 2.0+KG	12	6	50.0	6	50.0	10	9
Performance Rankings:							
Best FDA Approval Rate:	Gemini 2.0 Flash (61.8%)						
Best Novel Discovery:	GPT-3.5+KG (50.7%)						
Most Clinical Evidence:	GPT-3.5 Turbo (45 drugs)						

Drug-Target Network Analysis: We implemented a network analysis algorithm that identified drugs targeting shared molecular pathways. The system calculated pathway coverage scores for each drug candidate based on how many shared targets it affects.

Novel Target Discovery: The knowledge graph analysis revealed previously unexplored therapeutic opportunities by identifying shared molecular targets that are not currently addressed by existing treatments.

5.4.3 Key Findings from Implementation

Established Drug Classes: All three AI models consistently identified three major drug classes as top candidates:

SGLT2 Inhibitors: The models identified this class through analysis of glucose regulation pathways and cardiovascular protection mechanisms. Implementation showed these drugs target shared metabolic and cardiac pathways (Packer, 2023; Dyck et al., 2022).

GLP-1 Receptor Agonists: Analysis revealed dual mechanisms affecting both glucose control and cardiovascular outcomes. The implementation mapped these drugs to incretin pathways active in both conditions (Ma et al., 2021).

Statins: Models identified these through lipid metabolism pathway analysis and anti-inflammatory effects relevant to both heart disease and diabetes progression (Vogiatzi et al., 2017).

Novel Discoveries through Knowledge Graph Integration:

Zinc-Based Therapeutics: The knowledge graph integration identified zinc compounds through convergence analysis of seven shared genes: CRP, IL6, SLC22A3, SLC22A4, SLC22A5, TNF, and ADCY5. This discovery emerged from pathway analysis showing zinc's role in both insulin signaling and endothelial function (Chen et al., 2024; Xie et al., 2011).

Metformin Pathway Extensions: Beyond its known glucose-lowering effects, the system identified additional cardiovascular protective pathways through AMPK activation and inflammatory response modulation (Chaudhary and Kulkarni, 2024; Salvatore et al., 2020).

Natural Compounds: The implementation discovered resveratrol and berberine as candidates through their effects on SIRT1 and ADCY5 pathways, respectively (McCubrey et al., 2017).

The heart disease-diabetes case study demonstrates the practical application of our AI-driven drug repurposing approach. Through systematic analysis of this clinically significant comorbidity pair, we validated our methodology's ability to identify both established therapeutic options and novel repurposing opportunities.

Key Implementation Outcomes: The case study demonstrated the complementary strengths of different LLM models. GPT-3.5 provided comprehensive drug discovery with broad therapeutic coverage, Claude Sonnet 3.5 offered balanced and reliable predictions, while Gemini 2.0 Flash delivered efficient, high-confidence results.

Knowledge Graph Integration Value: The integration of molecular pathway data proved essential for discovering novel therapeutic targets like

zinc-based compounds and identifying extended mechanisms for established drugs like metformin. This approach revealed previously unexplored connections between disease pathways and potential therapeutic interventions.

Clinical Relevance: The identification of established drug classes (SGLT2 inhibitors, GLP-1 agonists, and statins) validates our approach's ability to recognize clinically accepted treatments, while the discovery of novel compounds demonstrates its potential for identifying new therapeutic opportunities in complex disease comorbidities.

6 Discussion

Our results highlight the substantial advantages of hybrid KG–LLM approaches over standalone methodologies in drug repurposing. The observed 56.4% reduction in candidate volume, paired with higher confidence scores, suggests that knowledge graph integration serves as an effective filter—removing low-quality predictions while retaining high-value therapeutic opportunities. This is particularly impactful for resource-constrained pharmaceutical pipelines, where downstream validation is costly and labor-intensive.

The system's ability to uncover previously unreported candidates (50.7% novel discovery rate) underscores the value of incorporating structured molecular knowledge. By capturing indirect relationships—such as shared targets or pathway overlap—KGs enable the discovery of non-obvious repurposing opportunities that LLMs alone might overlook. This is especially relevant for identifying drugs with pleiotropic effects or repositioning across diseases with overlapping etiology.

Furthermore, the high concordance between system predictions and established clinical guidelines validates the medical relevance of the outputs and supports the case for real-world adoption. The inclusion of interpretable confidence scores, grounded in both graph and text evidence, helps address critical barriers to clinical deployment: namely, transparency, factuality, and trust in AI-driven recommendations.

Our case study on heart disease–diabetes comorbidity demonstrates the practical utility of DrugEvolve in complex clinical decision-making contexts. The system successfully identified both established therapeutics (e.g., SGLT2 inhibitors, GLP-1 agonists) and novel candidates (e.g., zinc-based therapeutics), showcasing its dual capability for evidence-aligned prescribing and innovative hypothesis generation.

In addressing terminological variation—a persistent challenge in biomedical text mining—our semantic matching mechanism for disease normalization proves essential. By allowing configurable similarity thresholds, we maintain flexibility across clinical domains while preserving transparency and control in entity matching decisions.

Finally, our three-stage analytical pipeline—comprising LLM-only, KG-only, and hybrid configurations—offers a modular, extensible framework for methodological benchmarking and real-world adaptation. This design enables users to compare approaches systematically and facilitates the incorporation of emerging LLM architectures or updated KG resources, ensuring future compatibility and scalability.

7 Conclusion

We presented DrugEvolve, an open-source framework that integrates Knowledge Graphs with Large Language Models for interpretable, clinically relevant drug repurposing. By combining structured molecular knowledge with literature-based reasoning, DrugEvolve improves discovery precision, supports real-time decision-making, and achieves high concordance with clinical guidelines.

Looking ahead, we aim to enhance the clinical frontend, integrate multi-modal data, and pursue prospective validation studies to further bridge the gap between computational predictions and clinical application.

8 Ethical Considerations

This research involves the computational analysis of publicly available biomedical literature and knowledge graphs to support drug repurposing in clinical research workflows. The system is intended solely as a decision support tool to aid clinicians and researchers and is not a substitute for professional medical judgment or established clinical protocols. All data sources used are openly available and contain no personally identifiable or patient-specific information. Appropriate caution should be exercised when interpreting system outputs, particularly in high-stakes clinical contexts, and any use of the tool should be accompanied by expert oversight.

9 Limitations

Several limitations should be considered when interpreting these findings. LLMs are constrained by the temporal scope of their training data, potentially missing recent clinical developments, regulatory approvals, or emerging therapies. Knowledge graphs may lack full coverage of drug–target interactions, especially for novel mechanisms or recently characterized pathways. Literature-based analyses are also vulnerable to publication bias, which can overrepresent positive findings and inflate perceived therapeutic relevance. Moreover, AI-generated predictions require rigorous prospective clinical validation before clinical application. Future work should prioritize real-time literature integration, expanding KG coverage, and linking with electronic health records (EHRs) to improve clinical fidelity. Incorporating pharmacokinetic and safety profiling will further enhance the translational utility of AI-assisted decision support tools.

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Prompts

LLM only prompt

f"""Analyze the following collection of medical abstracts and extract relationships between diseases, comorbidities, and drugs. Only use the abstracts provided.

Your Task:

- Identify the **primary disease**: {primary_disease}. - Identify the **comorbidity** being analyzed: comorbidity. - List **drugs used to treat the primary disease**. - List **drugs used to treat the comorbidity**. - Identify **drugs that are effective for both conditions**.

n - For each drug that treats both conditions, provide: 1. A score from 1-10 on potential for repurposing (10 being highest)(If the drug is not explicitly mentioned in the literature, but is biologically plausible based on the knowledge graph, assign a score between 3–6 based on inferred mechanism.) 2. Evidence from literature for its dual efficacy 3. Mechanism of action explaining why it might work for both conditions 4. Potential molecular targets relevant to both conditions (if mentioned)

Output Format (JSON): {{ "primary_disease": "{primary_disease}", "comorbidity": "{comorbidity}", "drugs_primary_disease": ["Drug1", "Drug2"], "drugs_comorbidity": ["DrugA", "DrugB"], "shared_treatments": [{{ "drug": "Drug Name", "repurposing_score": 8, "primary_disease_treatment": true, "comorbidity_treatment": true, "evidence": "Brief evidence from literature", "mechanism_of_action": "Inhibits X receptor, reducing Y", "molecular_targets": ["target1", "target2"] }}], "explanation": "Overall explanation for drugs that are effective for both conditions." }}

Medical Abstracts: {full_text} """

LLM+KG prompt

prompt = f"""Analyze the following collection of medical abstracts and extract relationships between diseases, comorbidities, and drugs.

Primary Disease: {primary_disease} Comorbidity: {comorbidity}

{kg_context}

Your Task: - List drugs used to treat each disease. - Identify drugs that are effective for both conditions. - For each shared drug, provide: 1. Repurposing score (1–10)(If the drug is not explicitly mentioned in the literature, but is biologically plausible based on the knowledge graph, assign a score between 3–6 based on inferred mechanism.) 2. Evidence from abstracts 3. Mechanism of action 4. Mentioned or likely molecular targets

Output JSON Format: {{ "primary_disease": "...", "comorbidity": "...", "drugs_primary_disease": [...], "drugs_comorbidity": [...], "shared_treatments": [{{ "drug": "...", "repurposing_score": 7, "evidence": "...", "mechanism_of_action": "...", "molecular_targets": [...] }}], "explanation": "..." }}

Medical Abstracts: {full_text}